



WHICH FORECASTING MODEL PROVIDES THE
MOST ACCURATE PREDICTION FOR DAY-AHEAD
ELECTRICITY DEMAND IN NSW USING HISTORICAL
ELECTRICITY DEMAND AND WEATHER
VARIABLES?

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April 2026

SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENTS OF THE DEGREE OF
MASTER OF STATISTICS

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Abstract

In New South Wales, accurate forecasting of day-ahead electricity demand helps the power system plan and allocate power more effectively, and makes energy management more cost-effective. This study uses historical electricity demand and meteorological data to investigate which model can most accurately predict day-ahead electricity demand in New South Wales. In this study, four modelling approaches were compared: regression, time series, machine learning, and deep learning. The effects of lagged demand, temperature, day-of-week effects, month, and public holidays were also considered. The study was conducted on daily New South Wales electricity demand data sourced from the Australian Energy Market Operator, together with Bankstown temperature data as a representative weather variable for the state, and model performance was evaluated using standard forecasting metrics including MAE, RMSE, MAPE, and R^2 .

The results show that machine learning models achieved the strongest overall predictive performance. Among the standalone machine learning models, Gradient Boosting achieved the lowest MAE of 8,776 MWh and the lowest MAPE of 2.37%, while LightGBM achieved the lowest RMSE of 11,546 MWh and the highest R^2 of 0.9079. XGBoost also performed similarly, indicating that the three gradient boosting variants were all highly competitive after tuning. Among the time-series benchmarks, the SARIMAX(1, 1, 2) $\times$ (0, 0, 1)₇ model achieved an MAE of 11,325.73, an RMSE of 15,753.19, a MAPE of 2.99%, and an R^2 of 0.8280, indicating good forecasting performance for day-ahead electricity demand prediction. In the comparison of deep learning models, the ANN model outperformed the LSTM model across all evaluation metrics, suggesting that for a relatively limited daily dataset, a simpler architecture with well-engineered lag features may be more effective than a more complex sequence model.

Overall, the results indicate that machine learning models, particularly gradient boosting methods, are better able to capture lag persistence, calendar structure, and nonlinear demand patterns than regression, time series, and deep learning models for forecasting day-ahead electricity demand in New South Wales. These results provide practical evidence for selecting forecasting methods in electricity demand management and offer useful guidance for future improvements using richer weather inputs, broader exogenous variables, and more realistic forecast-time weather information.

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CHAPTER 1

Introduction

Electricity demand forecasting matters in a very practical sense. For power systems, day-ahead estimates are used to support scheduling and reserve decisions, so forecast errors can quickly become either reliability problems or unnecessary operating costs (Clements et al., 2016; Akhtar et al., 2023). If demand is underestimated, the system may not have enough reserve capacity. If it is overestimated, resources may be committed that are not actually needed (Clements et al., 2016). That is why short-term forecasting is still an important issue in power system operation rather than just a technical modelling task.

At the same time, this is not an easy forecasting problem. Electricity demand is influenced by several factors at once, including weather conditions, timing, seasonality, and holiday effects (Akhtar et al., 2023; Misiurek et al., 2025). Some of these relationships are fairly regular, while others are not. In practice, this means that a model may perform well in one setting but less well in another. Recent studies also show that the importance of accurate demand forecasting has grown with broader changes in power systems, especially with renewable integration, system planning, and increasingly data-intensive electricity markets (Ugbehe et al., 2025; Misiurek et al., 2025).

For that reason, it is difficult to justify choosing one modelling approach in advance. Previous research does not point to a single forecasting method that is best in every case (Vivas et al., 2020; Nti et al., 2020; Misiurek et al., 2025). Classical statistical models remain useful because their structure is easier to interpret, while machine learning and deep learning methods are often better suited to nonlinear patterns and more complex interactions (Vivas et al., 2020; Misiurek et al., 2025). There is also growing interest in hybrid approaches, which suggests that forecasting accuracy often depends on the prediction horizon, the characteristics of the dataset, and the inclusion of external variables such as weather and calendar information (Ugbehe et al., 2025; Vivas et al., 2020).

The NSW dataset used in this study gives a good reason to make that comparison explicit. It contains 4,084 daily observations from 8 January 2010 to 17 March 2021, with no missing values or duplicate rows in the variables retained for modelling. From the exploratory analysis, NSW electricity demand appears to move in a regular way over time. Today’s demand is usually similar to yesterday’s, and there is also a clear week-to-week pattern. Holiday effects are also visible, with average demand lower on holidays than on non-holidays, and there are noticeable differences across weekdays. By contrast, the temperature variables show only weak linear correlations with demand, even though they are strongly correlated with one

another. This makes the modelling task less straightforward, since weather appears relevant, but not in a purely linear way.

Given these patterns, this paper compares four groups of models: regression, time series, machine learning, and deep learning. The point is not simply to test different techniques for the sake of variety. It is to see which type of model works best for day-ahead electricity demand forecasting in New South Wales when historical demand, weather variables, and calendar information are considered together (Vivas et al., 2020; Chapagain et al., 2019; Clements et al., 2016). Model performance is evaluated using MAE, RMSE, MAPE, and R^2 so that the results can be compared on a consistent basis (Nti et al., 2020; Vivas et al., 2020). In this way, the study looks at the forecasting problem empirically rather than assuming beforehand which modelling family should dominate.

The remaining sections introduce the data and preprocessing steps, present the results from the different model groups, and then compare their performance before discussing the main findings and limitations.

CHAPTER 2

Literature Review

Day-ahead electricity demand forecasting is very closely tied to routine power system operation. In practice, system operators use these forecasts to decide how much generation and reserve capacity should be scheduled so that supply can match expected demand. So, when the forecast is too low, the system may face capacity pressure, but when it is too high, unnecessary operating costs can follow (Clements et al., 2016; Akhtar et al., 2023). For this reason, forecast accuracy matters not only statistically, but also in terms of reliability and operating efficiency.

The literature does not suggest that there is one forecasting method that works best in every setting. Recent reviews show that electricity demand forecasting has been approached through statistical models, machine learning and deep learning models. Hybrid frameworks that combine elements of both (Ugbehe et al., 2025; Misiurek et al., 2025; Vivas et al., 2020) have been experimented with as well. What matters finally is usually the forecasting horizon, the structure of the data, and the way external information, such as weather or calendar effects, is incorporated or used. In other words, model choice is context-dependent rather than universal.

Classical regression and time-series models still remain an important part of this literature. One reason is that they are easier to interpret, which makes them useful when the aim is not only prediction but also structured comparison across variables and model forms (Vivas et al., 2020). This is particularly relevant in day-ahead forecasting, where lagged demand and seasonal structure often carry substantial predictive information. Clements et al. (2016), for example, show that carefully specified time-series models can perform well in the Australian electricity market, especially when lagged load and related time effects are incorporated directly into the modelling framework.

At the same time, machine learning and deep learning methods have become more common because electricity demand is often shaped by nonlinear relationships and interacting effects. Reviews by Vivas et al. (2020) and Ugbehe et al. (2025) suggest that models which can flexibly incorporate exogenous variables often achieve strong forecasting performance. This is one reason why machine learning approaches are frequently used when demand depends on combinations of weather conditions, historical usage, and calendar patterns. Even so, the literature does not support the idea that more complex models are automatically better. In a short-term forecasting study for Thailand, Chapagain et al. (2019) found that multiple linear regression outperformed an artificial neural network. That result is useful here because it shows that forecasting accuracy cannot be assumed to increase simply with model complexity.

The choice of predictors is just as important as the choice of modelling method. Across the literature, three groups of variables appear most often: weather variables, historical demand, and calendar variables. Temperature is widely used because electricity consumption often changes with heating and cooling needs (Clements et al., 2016; Yildiz et al., 2018). In some studies, this effect is represented directly through temperature measures, while in others it is captured through variables such as cooling degree days (CDD) and heating degree days (HDD). Historical demand is also a standard predictor, since short-term electricity use is usually related to its recent past (Clements et al., 2016; Yildiz et al., 2018). In addition, calendar variables such as day of week, holidays, and season are commonly included because electricity demand tends to follow regular weekly and seasonal patterns (Wang et al., 2012; Yildiz et al., 2018).

Although these variables are widely recognised, the way they are handled is not always consistent across studies. Weather effects, in particular, are not always well captured by traditional linear methods, and earlier work often differs in how deeply weather variables are modelled (Alsmadi et al., 2025). Another issue is that many forecasting models rely heavily on stable historical patterns, which makes them more vulnerable during periods of disruption such as extreme weather events or the COVID-19 period (Fatema et al., 2023). This means that results reported in one context do not necessarily transfer cleanly to another, especially when climate conditions, consumption patterns, and feature construction differ.

For this project, one difficulty is that much of the literature is not directly comparable to the NSW case. Some papers are broad reviews, some study household-level demand, and others look at different national or regional markets. Many also test only one model family at a time. That makes it hard to know whether the reported performance comes from the model itself or from differences in the data, variables, and forecasting horizon. For a day-ahead NSW forecasting task, results from those studies are useful as background, but they do not provide a clean like-for-like comparison.

The gap addressed here is therefore fairly specific. Instead of introducing a new forecasting algorithm, this project compares regression, time-series, machine learning, and deep learning models within the same NSW setting, using the same forecasting horizon and a common set of demand and weather inputs. The aim is to see how these model families perform under comparable conditions, and to identify which approach is most useful for day-ahead electricity demand forecasting in this context. This kind of controlled comparison is valuable because it is more directly relevant to practical forecasting decisions than a discussion based only on separate results from different studies.

CHAPTER 3

Data and Exploratory Data Analysis

3.1 Study Objective, Dataset Construction and Overview

This chapter conducts a structured exploratory analysis of the dataset used for power demand forecasting. The analysis combines observed patterns in the data with descriptive statistics and visual evidence to examine the characteristics of the target variable `TOTAL_DEMAND`, thereby providing a foundation for subsequent modelling decisions.

The dataset contains 4,084 daily observations and 11 original variables, covering the period from 8 January 2010 to 17 March 2021. For EDA purposes, `LOCATION` and `HOLIDAY_NAME` were excluded, leaving nine variables for descriptive analysis, visualisation, and correlation assessment.

The target variable analysed in this study is a daily electricity demand measure constructed from the half-hourly `totaldemand_nsw` series. Specifically, for each calendar day, all half-hourly observations from 00:00 to 23:30 were aggregated and assigned to the corresponding date. For example, the daily electricity demand for 1 January 2010 was constructed from the half-hourly observations recorded between 2010-01-01 00:00 and 2010-01-01 23:30, while the value for 2 January 2010 was constructed from the observations recorded between 2010-01-02 00:00 and 2010-01-02 23:30. In this study, 00:00 was set as the daily aggregation boundary so that the demand series was aligned consistently with the daily weather, holiday, and weekday variables used in the merged dataset.

The temperature variables `T_MAX`, `T_MIN`, and `T_MEAN` denote the daily maximum temperature, daily minimum temperature, and daily mean temperature, respectively. We found that there are three missing days in temperature data, which are 2016-07-16, 2016-07-17 as well as 2016-07-18. And we used the mean temperature of that particular week to fill in for these three days. The variable `WEEKDAY` was derived from `DATE` and indicates the day of the week corresponding to each daily observation. The variable `IS_HOLIDAY` is a binary indicator of holiday status, where a value of 1 denotes that the day is a holiday and a value of 0 denotes that the day is not a holiday.

Before constructing the daily target variable, the original half-hourly demand series was checked for missing values. The results showed that the number of null values was zero, indicating that the raw demand series was complete and did not require imputation at this stage.

After the daily demand series was merged with the other explanatory variables, the lag features `DEMAND_LAG1` and `DEMAND_LAG7` were generated to capture short-term dependence and weekly patterns in electricity demand. Because these lagged variables require prior observations, the initial rows of the merged dataset contained

undefined lag values. These missing values were structural rather than arising from poor raw data quality. Therefore, the affected initial rows were removed before the exploratory analysis and subsequent modelling stages to ensure consistency in the analytical dataset.

Table 3.1: Dataset overview

Item	Description
Sample size	4,084 daily observations
Original variables	11
Variables used in EDA	9 (excluding LOCATION and HOLIDAY_NAME)
Date range	2010-01-08 to 2021-03-17
Forecast target	TOTAL_DEMAND (electricity demand)
Key candidate predictors	T_MAX, T_MIN, T_MEAN, WEEKDAY, IS_HOLIDAY, DEMAND_LAG1, DEMAND_LAG7

3.2 Data Quality Assessment

After eliminating the anomalies from the data, we processed and integrated our data together, and conducted a second check on this data set. This dataset has extremely high structural integrity. The check for missing values revealed that none of the nine variables retained had any missing values, with a missing rate of only 0%. The result of the check for duplicate rows also showed that the number of duplicate rows was zero. So now these data can be directly subjected to feature analysis and model preparation without the need for filling in missing values or removing duplicates.

The DATE field was parsed to `datetime` format and sorted in ascending order, ensuring that all subsequent time-series analyses are based on a properly ordered daily sequence.

Table 3.2: Data quality summary

Check	Result
Missing values	All EDA variables: 0 missing observations, 0% missing ratio
Duplicate rows	0 duplicated rows
Date field	Parsed to <code>datetime</code> and sorted chronologically
Overall integrity	No structural data-quality defects detected

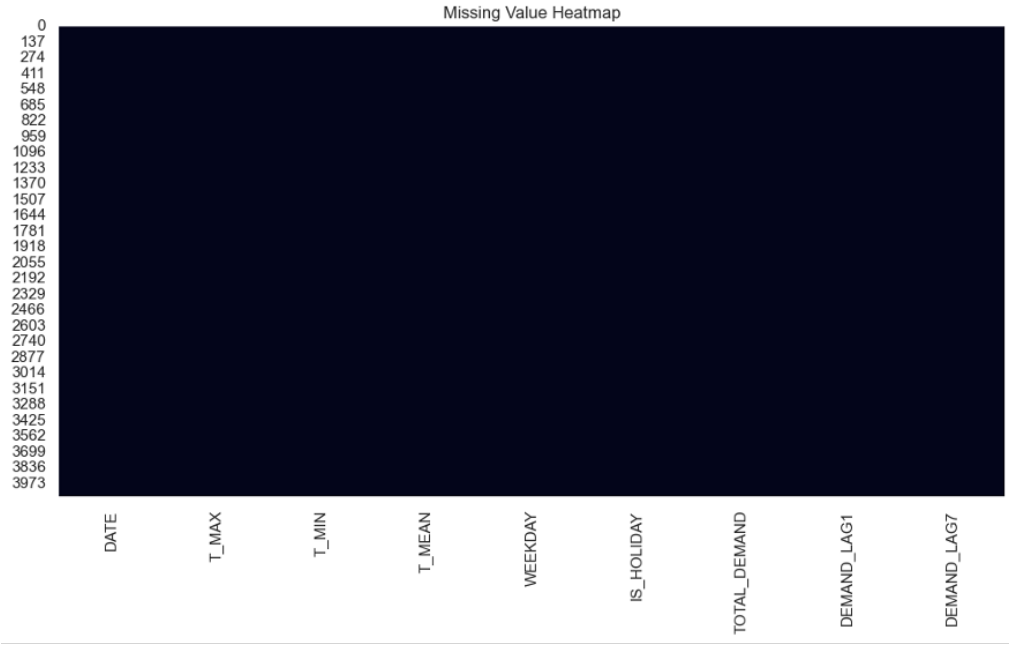


Figure 3.1: Missing-value heatmap confirming no missingness in the analysed variables.

3.3 Descriptive Statistics and Target Distribution

TOTAL_DEMAND has a mean of 389,412.42, a median of 386,998.83, a standard deviation of 38,359.37, a minimum of 290,102.22 as well as a maximum of 553,911.24. Its skewness is 0.329 and its kurtosis is approximately 0.015. This indicates that its distribution is roughly unimodal and the right tail is slightly extended. The target distribution is not severely skewed, but there are still some peak days with high demand that can be observed, which may have an impact on the distribution of the regression error.

This pattern indicates that this series of data is suitable for direct modeling without the need for drastic transformation processing. This is the default choice. However, during the model evaluation process, the peak demand data still needs to be closely monitored.

Table 3.3: Summary statistics for **TOTAL_DEMAND**

Statistic	Value
Mean	389,412.42
Median	386,998.83
Standard deviation	38,359.37
Minimum	290,102.22
Maximum	553,911.24
Skewness	0.329
Kurtosis	0.015

Table 3.4: Selected descriptive statistics for key numeric variables

Variable	Mean	Std. Dev.	Min	Max
T_MAX	22.91	5.05	11.70	44.70
T_MIN	12.49	5.29	−1.30	25.60
T_MEAN	17.40	4.69	4.94	33.31
TOTAL_DEMAND	389,412.42	38,359.37	290,102.22	553,911.24
DEMAND_LAG1	389,413.19	38,346.33	290,102.22	553,911.24
DEMAND_LAG7	389,479.23	38,346.82	290,102.22	553,911.24

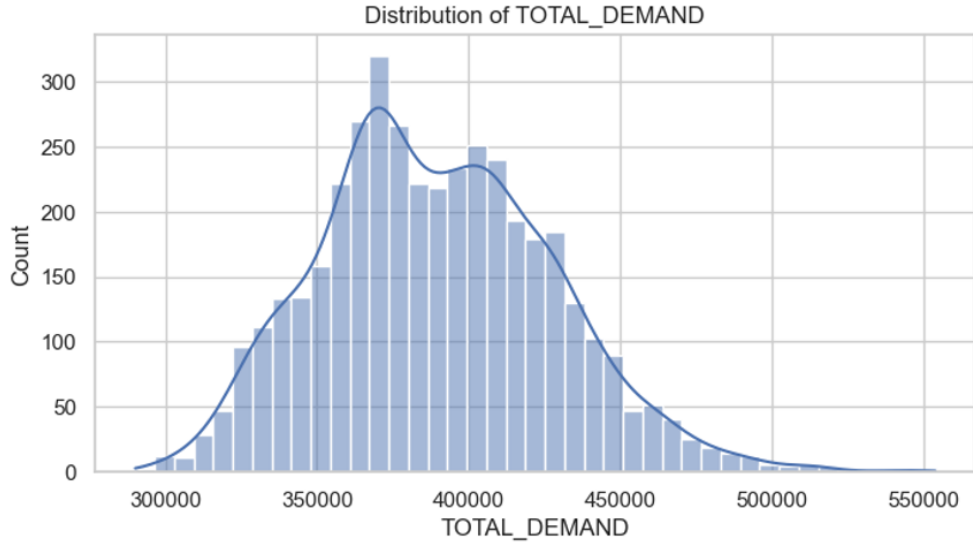


Figure 3.2: Distribution and boxplot of TOTAL_DEMAND.

3.4 Temporal Trend and Seasonality

The daily time-series plot graph clearly shows a significant temporal correlation and recurring fluctuations in TOTAL_DEMAND. In the long term, the demand for the early stages of the sample data is relatively high, while it decreases in the later stages. However, overall, the demand fluctuations remain quite significant.

At the seasonal level, the recurring peaks and troughs indicate a regular cyclical behavior, which is related to the influence of weather conditions, calendar effects, and habitual consumption patterns on demand. The annual box plot shows the differences in the central tendency and dispersion between years, while the monthly average demand curve confirms the obvious monthly seasonal components.

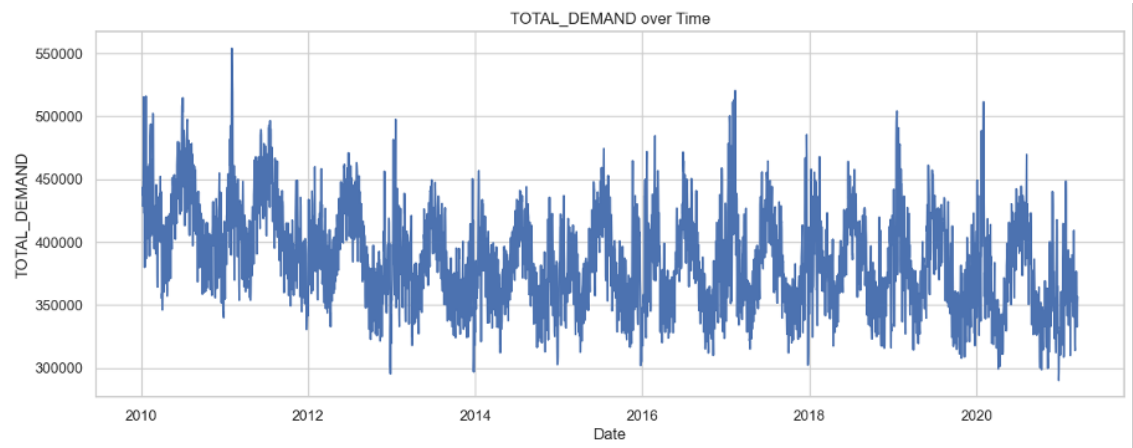


Figure 3.3: Daily trajectory of TOTAL_DEMAND over the sample period.

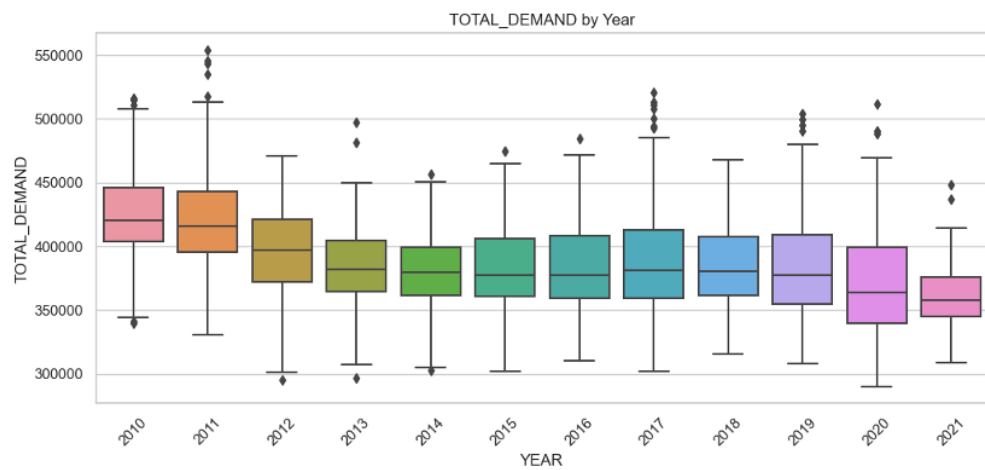


Figure 3.4: Distribution of TOTAL_DEMAND by year.

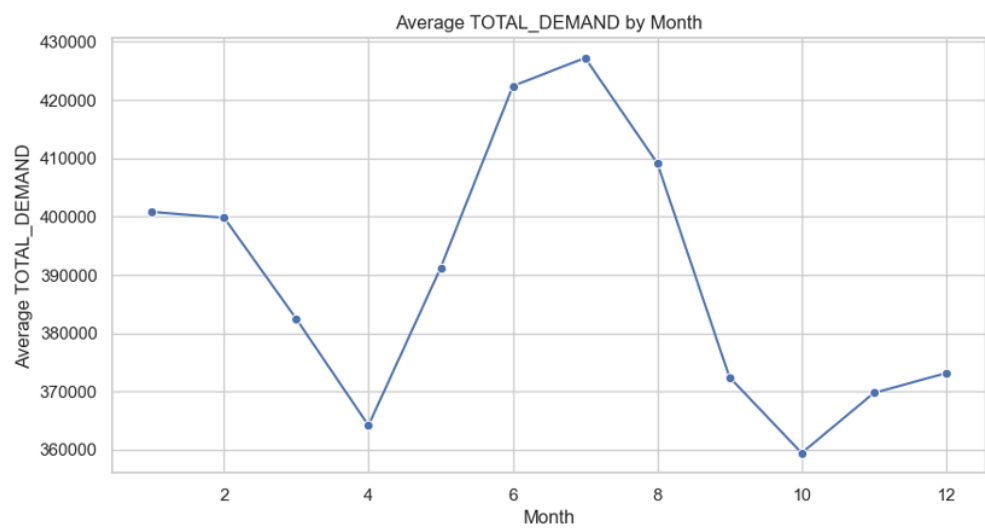


Figure 3.5: Average TOTAL_DEMAND aggregated by month.

3.5 Calendar Effects: Weekday and Holiday Patterns

The distribution of `WEEKDAY` is nearly balanced, with each coded category containing approximately 583–584 observations. Despite this balance, average demand differs materially across weekday codes: `WEEKDAY = 2` records the highest average demand at 401,191.72, whereas `WEEKDAY = 7` records the lowest at 359,915.08. This confirms that intra-week behavioural patterns are economically meaningful for demand forecasting.

The holiday effect is even more pronounced. Average demand on non-holidays is 390,922.34, compared with 343,487.67 on holidays—a reduction of approximately 47,434.67, or roughly 12.1%. Although holiday observations constitute only 130 records (3.18% of the sample), both the direction and magnitude of the effect are unambiguous.

Table 3.5: Average `TOTAL_DEMAND` by weekday code

Weekday code	Average <code>TOTAL_DEMAND</code>
2	401,191.72
4	400,833.81
3	399,591.26
5	398,786.43
1	395,182.85
6	370,332.06
7	359,915.08

Table 3.6: Average `TOTAL_DEMAND` on holidays vs. non-holidays

Group	Average <code>TOTAL_DEMAND</code>
Non-holiday	390,922.34
Holiday	343,487.67

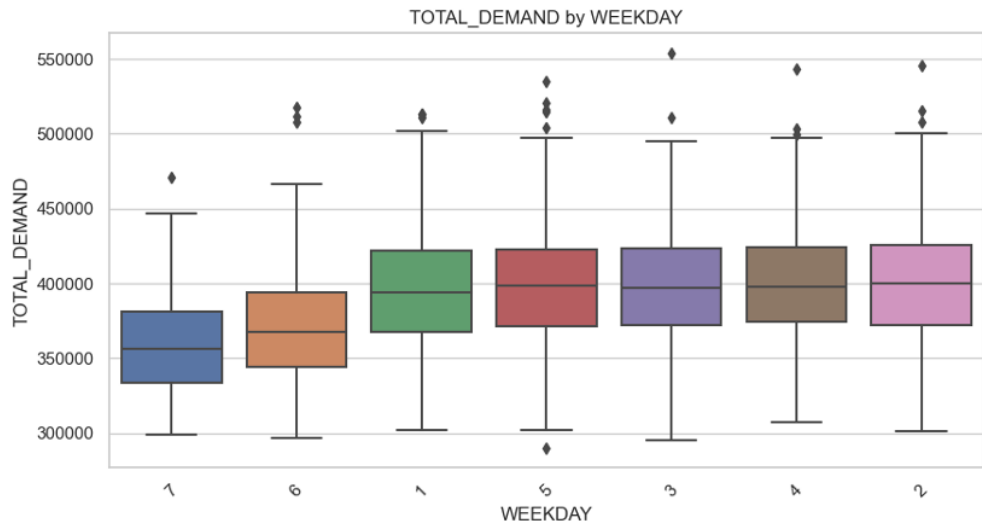


Figure 3.6: Effect of `WEEKDAY` on `TOTAL_DEMAND`.

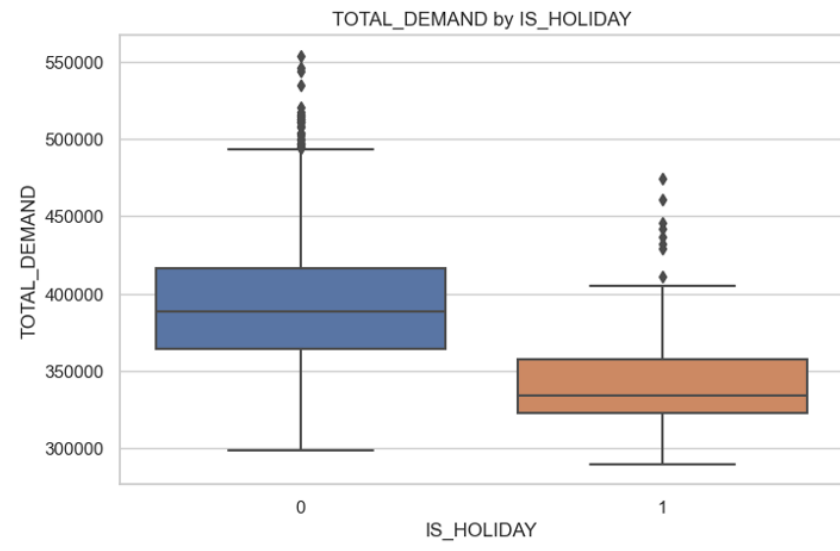


Figure 3.7: Difference in `TOTAL_DEMAND` between holidays and non-holidays.

3.6 Relationships Between Predictors and the Target

The correlation analysis shows that the lag features are by far the strongest linear correlates of `TOTAL_DEMAND`: 0.7994 for `DEMAND_LAG1` and 0.7154 for `DEMAND_LAG7`. These values substantially exceed those of the temperature or calendar variables, indicating that short-run persistence and weekly recurrence are core properties of the demand process.

Temperature variables exhibit weak negative linear correlations with the target: -0.0998 for `T_MAX`, -0.1548 for `T_MIN`, and -0.1641 for `T_MEAN`. This does not mean that temperature can be disregarded; a more reasonable explanation is that the relationship between temperature and demand is not linear, and it also interacts with seasonal and calendar factors.

`WEEKDAY` and `IS_HOLIDAY` correlate with the target at -0.3134 and -0.2171 , respectively. Although smaller than the lag correlations, these values are sufficient to justify explicit retention in downstream modelling.

Table 3.7: Summary of predictor relationships with `TOTAL_DEMAND`

Variable	Type	Missing count	Missing ratio (%)	Correlation with target
<code>DEMAND_LAG1</code>	float64	0	0.0	+0.7994
<code>DEMAND_LAG7</code>	float64	0	0.0	+0.7154
<code>T_MAX</code>	float64	0	0.0	-0.0998
<code>T_MIN</code>	float64	0	0.0	-0.1548
<code>T_MEAN</code>	float64	0	0.0	-0.1641
<code>IS_HOLIDAY</code>	int64	0	0.0	-0.2171
<code>WEEKDAY</code>	int64	0	0.0	-0.3134

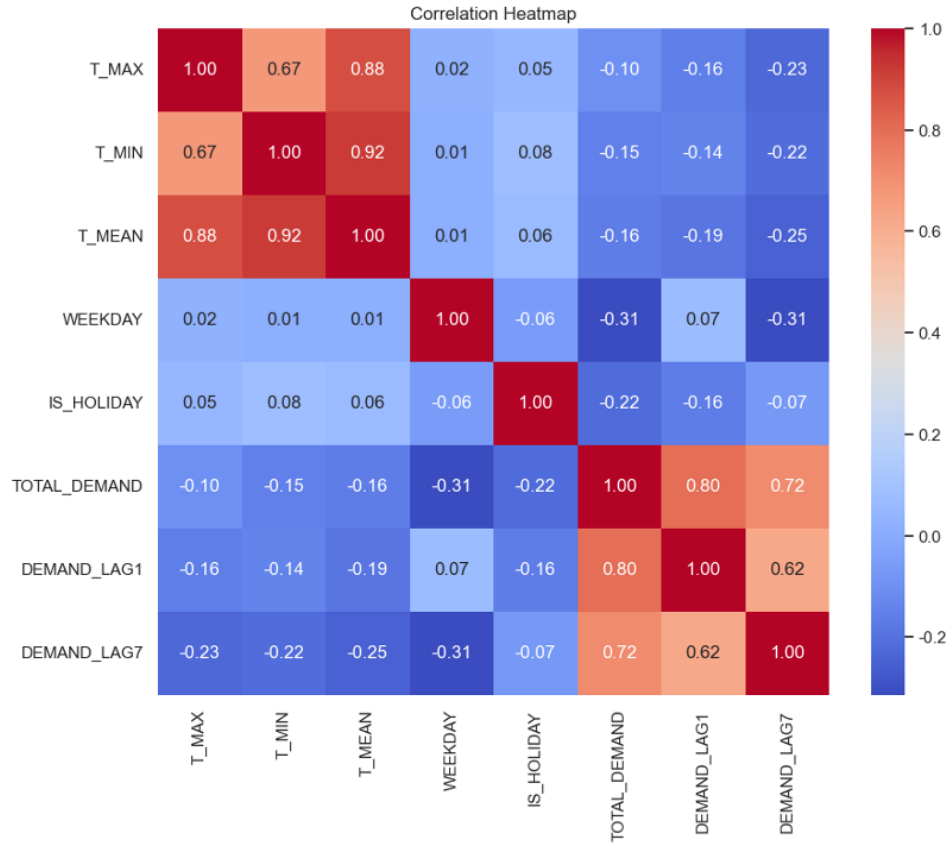


Figure 3.8: Correlation heatmap across numeric variables.

3.7 Lag Structure and Autocorrelation

The scatterplot of `TOTAL_DEMAND` against `DEMAND_LAG1` forms a tight positive band, confirming strong adjacent-day persistence. The relationship with `DEMAND_LAG7` is a little bit weaker, but it is still obvious. This is consistent with the weekly recurring demand cycle.

The short-lag ACF indicates that `TOTAL_DEMAND` remains significantly positively autocorrelated across several low-order lags, with the strongest dependence at small displacements, which implies strong short-term persistence. The long-lag autocorrelation plot further shows an oscillatory decay pattern, indicating sustained seasonal or cyclical structure beyond the short horizon.

These findings have direct modeling implications: lagged inputs, rolling statistics, seasonal indicators, or time-aware forecasting architectures should be prioritised. A model that ignores the temporal structure and relies only on the same-day weather and calendar inputs would leave a substantial portion of the predictive signal unexploited.

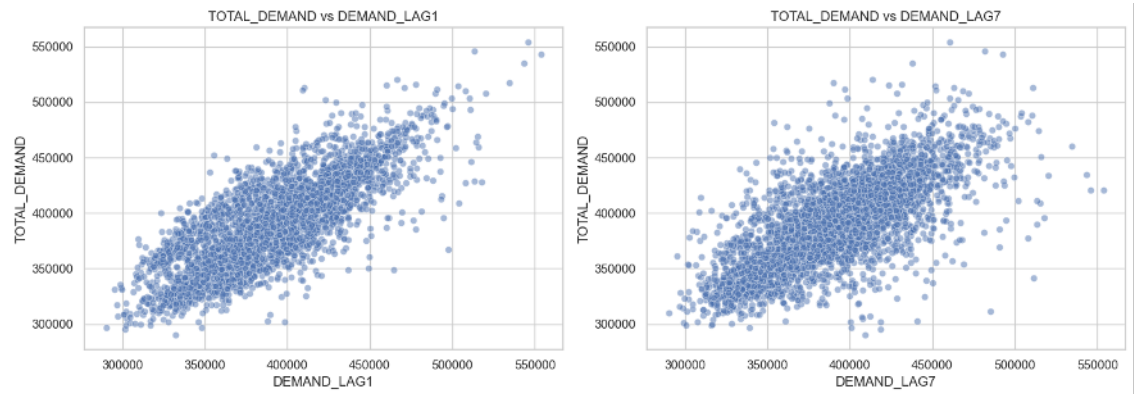


Figure 3.9: Scatterplots of `TOTAL_DEMAND` against `DEMAND_LAG1` (left) and `DEMAND_LAG7` (right).

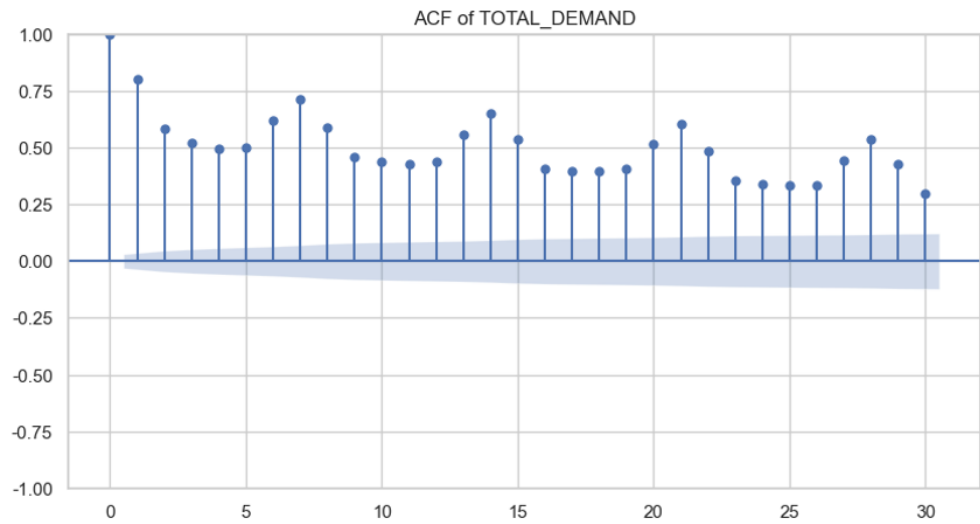


Figure 3.10: Short-lag autocorrelation function (ACF) of `TOTAL_DEMAND`.

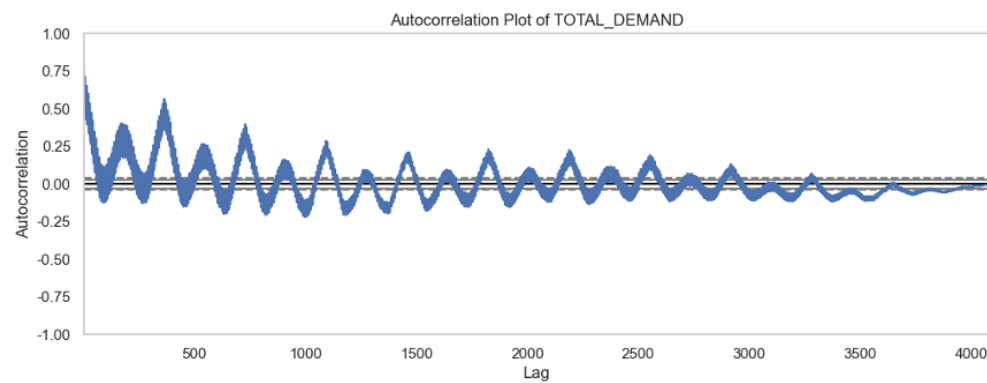


Figure 3.11: Long-range autocorrelation plot of `TOTAL_DEMAND`.

3.8 Temperature Multicollinearity and Outliers

The three temperature measures are strongly inter-correlated. The Pearson correlations are 0.8786 between T_MAX and T_MEAN, 0.9189 between T_MIN and T_MEAN, and 0.6720 between T_MAX and T_MIN. This indicates substantial multicollinearity, which may cause instability in coefficient-sensitive models such as ordinary least squares. Therefore, regularization processing or feature selection operations may be necessary.

By contrast, the pairwise correlations of all three temperature variables with TOTAL_DEMAND are weakly negative (-0.155 , -0.100 , and -0.164 , respectively). These lower values should not be interpreted as evidence suggesting that temperature is unimportant; they are more likely to be caused by nonlinear effects, seasonal variations, or interactions with other time variables. Therefore, temperature-related predictors should be retained in the prediction framework, and perhaps supplemented through nonlinear transformations, or incorporated into machine learning models that can capture complex, high-order relationships.

Outlier identification using the interquartile range (IQR) rule reveals modest outlier ratios overall. TOTAL_DEMAND, DEMAND_LAG1, and DEMAND_LAG7 each exhibit an outlier ratio of approximately 0.66%, while T_MAX shows the highest ratio at around 1.03%. Since these observational results may be related to extreme weather conditions or periods of operational stress, they should be handled with caution rather than simply being discarded.

Table 3.8: Correlations among temperature variables

	T_MAX	T_MIN	T_MEAN
T_MAX	1.0000	0.6720	0.8786
T_MIN	0.6720	1.0000	0.9189
T_MEAN	0.8786	0.9189	1.0000

Table 3.9: Outlier summary based on the IQR rule

Variable	Outlier count	Outlier ratio (%)
IS_HOLIDAY	130	3.183
T_MAX	42	1.028
TOTAL_DEMAND	27	0.661
DEMAND_LAG1	27	0.661
DEMAND_LAG7	27	0.661
T_MEAN	1	0.024
T_MIN	0	0.000
WEEKDAY	0	0.000

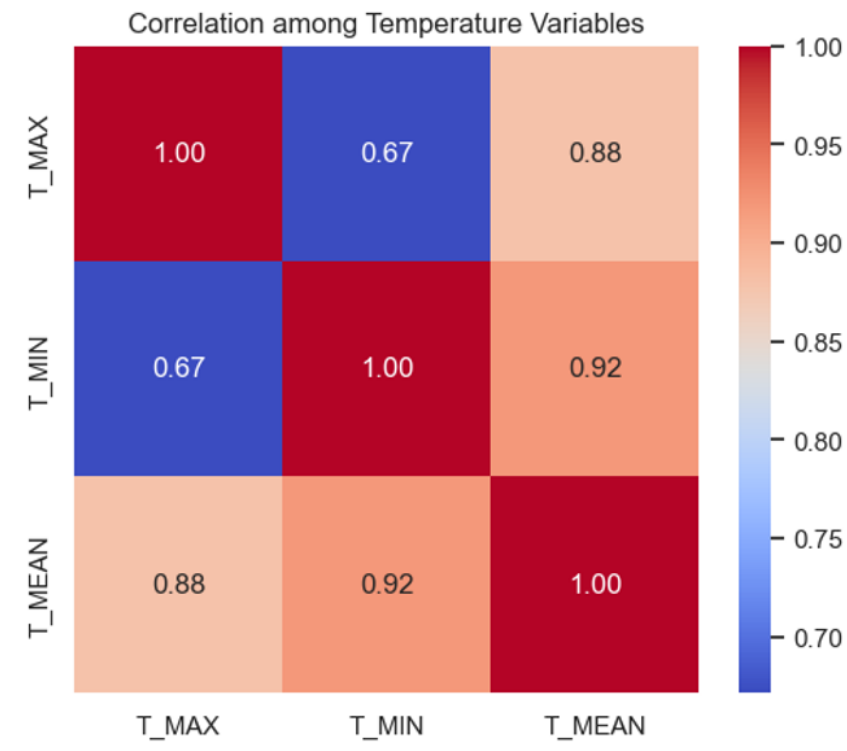


Figure 3.12: Correlation heatmap among temperature variables.

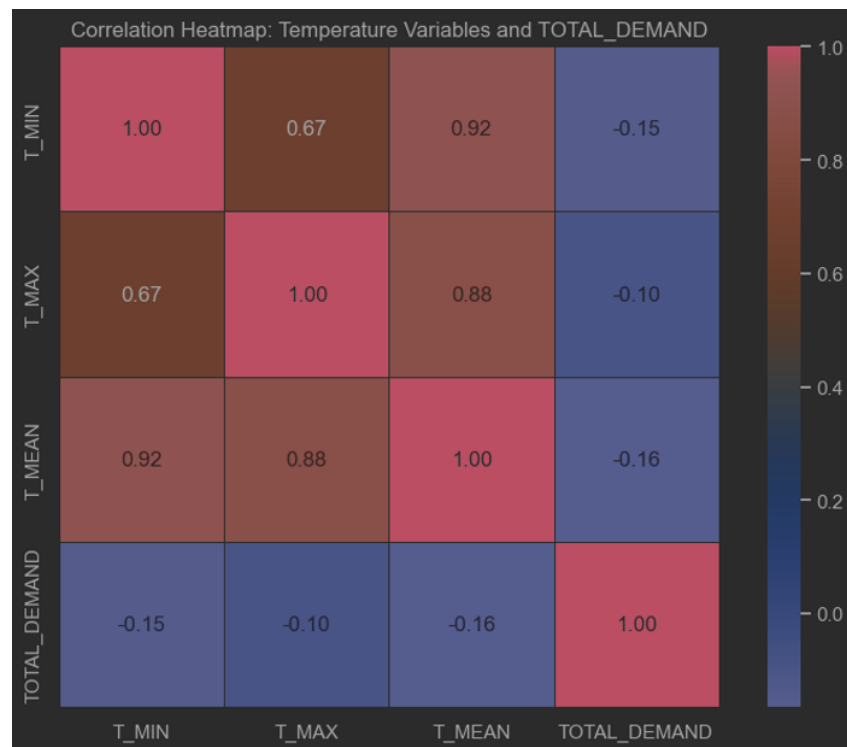


Figure 3.13: Correlation heatmap among temperature variables and TOTAL_DEMAND.

3.9 Interaction Patterns and Modelling Implications

The joint month–holiday view indicates that the magnitude of the holiday effect varies across months, suggesting that holiday impacts are seasonally modulated rather than constant. Likewise, the weekday–holiday cross-tabulation shows that holiday demand remains lower than non-holiday demand for every weekday code, confirming a robust holiday effect across the weekly cycle.

Taken together, the EDA supports four key modelling implications. Firstly, given the strong linear correlation between the lagged variable and the target variable, it should be regarded as the primary predictive factor. Secondly, the temperature effect may be nonlinear, and there is a problem of collinearity among the features. Therefore, it is recommended to adopt flexible feature engineering or regularization modeling methods. Thirdly, `WEEKDAY` and `IS_HOLIDAY` should be retained as primary calendar covariates. At the fourth place, rare peak-demand observations should be addressed through robust evaluation strategies rather than mechanical deletion.

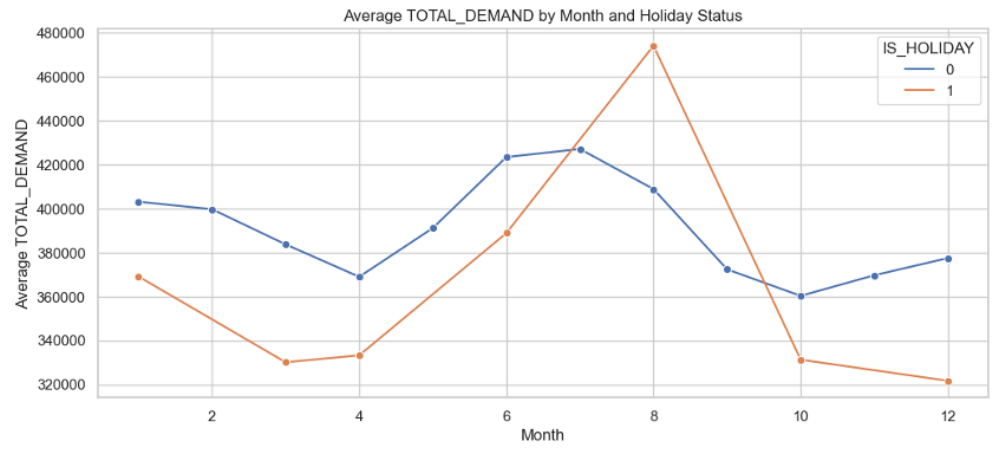


Figure 3.14: Joint effect of month and holiday status on `TOTAL_DEMAND`.

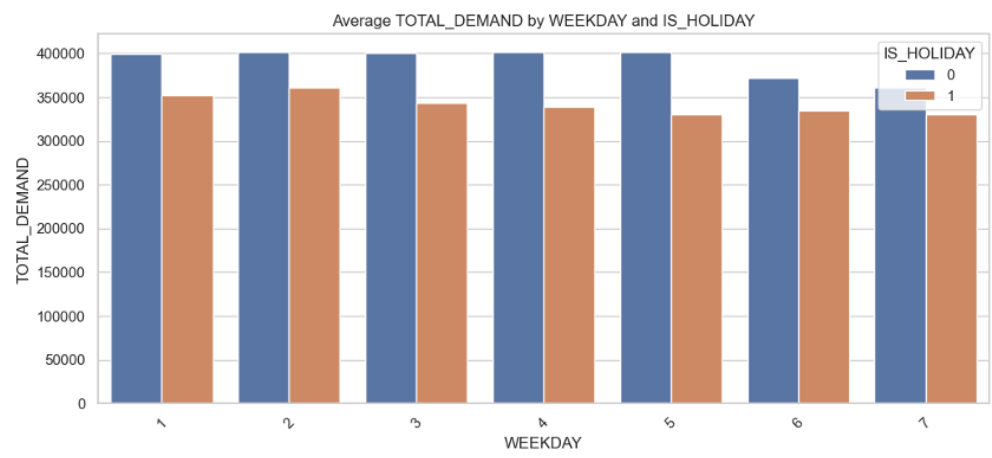


Figure 3.15: Joint effect of weekday code and holiday status on `TOTAL_DEMAND`.

CHAPTER 4

Methodology

4.1 Regression Models

4.1.1 Model Selection

This study formulates daily electricity demand forecasting as a supervised regression problem in which `TOTAL_DEMAND` is the response variable and temperature, calendar, and lagged-demand features are used as predictors. This design is in line with the relevant literature in the field of power forecasting. The literature repeatedly points out that weather parameters, historical demand, and calendar effects are the main sources of information for short-term load forecasting.

The methodological setting is also anchored in the Australian day-ahead forecasting context. Clements et al. (2016) show, using Queensland data from the National Electricity Market, that strong day-ahead performance can be achieved when lagged information is combined with seasonal structure, and they explicitly identify the effective use of lagged information and interaction with recurring temporal patterns as the key ingredient of forecast success.

Rather than assuming that one modelling family is universally superior, the present study compares interpretable linear baselines with a flexible non-linear ensemble model. This comparative design is supported by the review literature: Vivas et al. (2020) emphasise that forecasting performance depends on forecasting horizon, data properties, and the use of exogenous variability, while Nti et al. (2020) note that no single validity index or modelling approach dominates every forecasting setting.

The target variable is `TOTAL_DEMAND`. The explanatory variables come from three structural categories: temperature variables, calendar variables, and lagged demand variables. Specifically, the raw candidate predictors are `T_MAX`, `T_MIN`, `T_MEAN`, `WEEKDAY`, `IS_HOLIDAY`, `DEMAND_LAG1`, and `DEMAND_LAG7`.

The selection of temperature, calendar, and historical load characteristics is largely based on previous literature. Nti et al. (2020) report that a large share of electricity demand forecasting studies uses weather-related parameters and historical consumption information, and they identify RMSE and MAPE as the most common evaluation metrics. Ugbehe et al. (2025) further classify electricity demand forecasting methods into statistical, machine learning/artificial intelligence, and hybrid approaches, while emphasising that the quality of prediction largely depends on the variables used and the verification strategy.

The lagged demand terms are theoretically central. `DEMAND_LAG1` captures short-run persistence between adjacent days, while `DEMAND_LAG7` captures weekly recurrence in human activity and electricity usage. This treatment aligns directly with

Clements et al. (2016), whose Australian day-ahead study attributes strong predictive performance to the effective use of lagged information and seasonal dependence.

The treatment of temperature variables has been adjusted. Because `T_MAX`, `T_MIN`, and `T_MEAN` are naturally highly correlated, using all three simultaneously in linear regression may inflate coefficient variance and weaken interpretability. Accordingly, the linear-family models—Linear Regression, Ridge, and Lasso—are estimated with `T_MAX` and `T_MIN` but exclude `T_MEAN` (12 predictors in total). The tree-based model retains all three temperature variables (13 predictors) because its predictive mechanism is less sensitive to coefficient instability and can exploit redundant but informative splits more flexibly. This segmented design not only avoids unnecessarily restricting the non-linear model but also enhances the interpretability and stability of the linear benchmark.

In addition, temperature outliers identified by the IQR rule were not removed mechanically. Their overall ratios were modest, and such observations may reflect genuine extreme weather conditions rather than measurement error. In the context of electricity demand forecasting, extreme temperatures are substantively meaningful because they may coincide with unusually high or stressed demand conditions. Therefore, temperature outliers were retained unless there was clear evidence of data inconsistency.

The four selected models serve different purposes. Linear Regression is included as the most transparent benchmark and provides a direct additive baseline. Ridge Regression and Lasso Regression are included because the exploratory analysis suggests that the temperature variables are correlated, so regularised linear models are useful for improving coefficient stability and reducing redundancy. Random Forest is included because electricity demand is determined by factors such as lag, calendar repetition, and weather conditions, and these factors are unlikely to interact with each other in a strictly linear and cumulative manner. Therefore, Random Forest provides a suitable non-linear benchmark within the regression framework.

4.1.2 Model Specification

Linear Regression. Linear regression models the conditional mean of electricity demand as a linear function of the predictors. Its role in this study is to provide a transparent baseline and a direct estimate of additive effects:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \varepsilon_i \quad (4.1.1)$$

where y_i denotes the electricity demand observed for observation i ; x_{ij} denotes the value of predictor j for observation i ; p is the number of predictors; and n is the total number of observations. The parameter β_0 is the intercept, while β_j is the coefficient for predictor j , representing the expected change in demand for a one-unit increase in x_{ij} , holding other variables constant. The term ε_i is the random error term capturing variation not explained by the predictors.

Ridge Regression. Ridge regression extends linear regression by adding an L2 penalty to the loss function, which is useful when predictors are correlated:

$$\min_{\beta} \left[\sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p \beta_j^2 \right] \quad (4.1.2)$$

where the first term is the residual sum of squares and the second term is the ridge penalty. The tuning parameter $\lambda \geq 0$ controls regularisation strength: when $\lambda = 0$ the model reduces to ordinary least squares; as λ increases, the coefficients are shrunk more strongly towards zero. The penalty is applied to β_j , $j = 1, \dots, p$, but not to the intercept β_0 .

Lasso Regression. Lasso regression adds an L1 penalty and can shrink some coefficients exactly to zero, enabling implicit feature selection:

$$\min_{\boldsymbol{\beta}} \left[\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p |\beta_j| \right] \quad (4.1.3)$$

where $|\beta_j|$ is the absolute value of coefficient β_j . As λ increases, more coefficients are pushed towards zero, and some become exactly zero, making Lasso useful when the predictor set contains redundant or weakly informative variables.

Random Forest Regression. Random forest is an ensemble model composed of regression trees constructed based on bootstrap sampling. During each split, random feature selection is performed. The final prediction is the average of many decorrelated trees, reducing variance and capturing non-linear interactions:

$$\hat{y}(\mathbf{x}) = \frac{1}{B} \sum_{b=1}^B T_b(\mathbf{x}) \quad (4.1.4)$$

where B is the total number of trees and $T_b(\mathbf{x})$ is the prediction of the b -th tree for input vector $\mathbf{x}$. Because each tree is trained on a bootstrap sample and uses random feature selection during splitting, the trees are deliberately decorrelated, so averaging reduces variance and improves predictive robustness.

The inclusion of both linear and non-linear models is methodologically justified by the literature. Vivas et al. (2020) show that forecasting accuracy depends on the forecasting horizon and the use of exogenous variability, while Nti et al. (2020) emphasise that no single method is scientifically superior in all situations. Accordingly, the present study does not assume a priori that machine learning will outperform regression-based baselines, but tests that proposition empirically on the NSW dataset.

4.1.3 Training Strategy

Because the observations are temporally ordered, the dataset is first sorted by **DATE** and then split chronologically into training and testing subsets using an 80:20 ratio. The earlier 80% of observations are used for model estimation and the later 20% are reserved for out-of-sample testing. A random split is intentionally avoided because it would mix future information into the training stage and would not represent a genuine forecasting environment.

This time-sequential assessment design is conceptually consistent with the literature on forecasting. Power demand forecasting is influenced by lead time, time dependence, and seasonality. Therefore, an effective test should simulate the prediction of an unseen future period rather than predicting random observations drawn from the same time mixture.

In the revised workflow, two feature sets are used. Linear-family models are trained with 12 predictors, excluding `T_MEAN`, while Random Forest is trained with 13 predictors and retains the full temperature set.

To strengthen model calibration, hyperparameter tuning was conducted for all four regression models using `GridSearchCV`. Because ordinary random cross-validation would break the chronological structure of the data, the tuning process used `TimeSeriesSplit` within the training sample. This design preserves temporal ordering across folds and better reflects the forecasting setting than standard shuffled validation.

For each model, a finite parameter grid was defined and candidate specifications were evaluated using time-series-aware cross-validation. The selection criterion was the minimisation of cross-validated RMSE. After the optimal hyperparameter combination had been identified, the model was re-fitted on the full training set and then evaluated on the independent test set. This design separates parameter selection from final out-of-sample assessment and allows a transparent and fair comparison among the tuned models.

The grid search produced the following best-performing parameter combinations:

- **Random Forest:** `max_depth = None, min_samples_leaf = 1, min_samples_split = 2, n_estimators = 400`
- **Lasso:** `alpha = 1.0, fit_intercept = True`
- **Ridge:** `alpha = 10.0, fit_intercept = True`
- **Linear Regression:** `fit_intercept = True`

Model performance is evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and the coefficient of determination (R^2). RMSE and MAPE are especially well established in the forecasting literature: Nti et al. (2020) report RMSE as the most frequently used error metric and MAPE as the next most common, while Clements et al. (2016) use MAPE as the principal criterion in Australian day-ahead load forecasting.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (4.1.5)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (4.1.6)$$

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (4.1.7)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (4.1.8)$$

4.1.4 Results

Table 4.1 summarises the out-of-sample performance of the four tuned regression models. Random Forest is the best-performing model on every metric, while the three linear-family models remain close to one another and substantially below the tree-based model in predictive accuracy.

Table 4.1: Out-of-sample performance of the tuned regression models on the NSW electricity demand test set. Lower MAE, RMSE, and MAPE indicate better performance; higher R^2 indicates better fit. The column “Feat.” records the number of predictors used.

Model	MAE	RMSE	R^2	MAPE (%)	Feat.
Random Forest	10,499.60	13,935.97	0.8659	2.8435	13
Lasso	14,139.75	18,580.71	0.7616	3.7319	12
Ridge	14,140.70	18,585.18	0.7614	3.7317	12
Linear Regression	14,192.85	18,615.53	0.7607	3.7479	12

The strongest model, Random Forest, achieves an R^2 of 0.8659, an RMSE of 13,935.97, and a MAPE of 2.8435%, indicating that the chosen feature set contains substantial forecasting information for daily electricity demand. The three linear models produce very similar error levels: Ridge and Lasso are almost indistinguishable, and ordinary Linear Regression is only marginally weaker. Even after removing `T_MEAN` from the linear specifications, the linear models remain tightly clustered, suggesting that the main performance constraint is not only coefficient instability but also the inability of a purely linear functional form to capture threshold effects and interactions.

Theoretically, the advantage of the Random Forest model lies in its rationality, as it is based on the structure of daily electricity demand. Electricity demand is determined by factors such as lag, calendar repetition, and weather conditions, and these factors are unlikely to interact with each other in a strictly linear and cumulative manner. Temperature effects often intensify at extreme levels, and holiday conditions can alter how lagged demand maps into the following day. A tree-based ensemble model is better positioned than a linear model to represent such threshold behaviour and higher-order interactions.

The comparison between the linear-family models is also informative. Once `T_MEAN` is removed from the linear specifications, the almost identical results of Linear Regression, Ridge, and Lasso suggest that multicollinearity was a valid modelling concern but not the principal explanation for the linear models’ weaker performance. In other words, regularisation improves methodological cleanliness, yet it does not close the performance gap because the remaining structure to be learned is still predominantly non-linear.

The final tuned results further support this explanation. The performance of Random Forest is clearly better than that of the linear models, which confirms that non-linear modeling is valuable for the current dataset. The reasonable explanation is that the daily demand sequence in New South Wales contains interaction and

threshold effects, and these effects are better captured by the ensemble structure of Random Forest than by additive linear models.

These findings are generally consistent with the existing literature, but the results should not be exaggerated. Vivas et al. (2020) show that forecast performance depends on horizon and exogenous variability, while Ugbehe et al. (2025) describe the growing relevance of ML/AI and hybrid methods. At the same time, Clements et al. (2016) demonstrate that carefully specified regression models can remain highly competitive in day-ahead Australian forecasting. Therefore, the defensible conclusion is not that machine learning is always superior, but that under the present NSW daily dataset, feature design, and validation setup, Random Forest captures the available structure more effectively than the tested linear baselines.

4.1.5 Limitations

Before considering the current research results as the final conclusion, some limitations should be acknowledged. Firstly, the current evidence is based on a dataset and an 80:20 proportion divided in chronological order. To obtain more robust evidence, methods based on rolling start points or forward validation need to be conducted in multiple prediction windows. Secondly, the feature space is still limited. The model has not incorporated humidity, wind conditions, extreme weather indicators, broader economic drivers, or more comprehensive holiday encodings. Thirdly, the current exercise only assesses the accuracy of point predictions, without estimating the prediction intervals or uncertainty ranges, which are crucial for actual power system planning. Fourthly, the comparison results are derived based on moderate but not exhaustive adjustments of hyperparameters. Therefore, the reported rankings represent the relative performance of the optimized implementation methods, rather than the absolute best performance that each algorithm can achieve. Finally, there are differences in the interpretability of the model. The model that performed the best is still an ensemble method. And its internal logic is not as clear and understandable as the coefficient-based regression methods. For a final thesis version, further analysis should be strengthened through methods such as residual diagnosis, feature importance analysis as well as robustness testing. Where feasible, interpretable machine learning tools such as partial dependence plots or SHAP-based explanations should be used.

4.2 Time Series Modelling Approach

4.2.1 Model Selection

Electricity demand is suitable for time series analysis because the data are recorded over time and the current level of demand is often related to earlier observations. In day-ahead forecasting, electricity demand usually shows repeated daily, weekly, and seasonal patterns. Because of these features, time series models are useful for analyzing how the demand changes over time. Previous research has found that time series methods work well in short-term electricity demand forecasting, especially when the demand series has stable and repeated patterns (Akhtar et al., 2023; Czapaj et al., 2022).

For this paper, SARIMAX was chosen as the main time series model because it can capture both the seasonal patterns of electricity demand and the influence of external variables. Although ARIMA is useful because it can easily capture

the relationship between current and past demand values, it is less suitable when seasonality is an important feature of the series. In day-ahead electricity demand forecasting such patterns are common, so a seasonal specification is more appropriate. For this paper, electricity demand is influenced not only by its previous values, but also by temperature, holidays, and differences across weekdays. Therefore, SARIMAX is more appropriate for this study, since it allows these factors to be included in one model. This is also consistent with previous studies showing that autoregressive and ARIMA-based methods remain useful in short-term electricity demand forecasting, including applications in Australia (Al-Musaylh et al., 2018; Czapaj et al., 2022).

To provide a clearer comparison between different time-series frameworks, this study also includes a SARIMA model as a benchmark. SARIMA can capture autoregressive behavior, differencing effects, moving average structure, and seasonal patterns in electricity demand, but it does not include exogenous variables. By comparing SARIMA with SARIMAX, it is possible to assess whether adding external variables such as temperature, holiday effects, and day-of-week indicators improves day-ahead forecasting performance.

4.2.2 Model Specification

The SARIMAX model was used in this study to predict day-ahead electricity demand by combining both historical electricity demand values and external variables. In SARIMAX model, electricity demand was treated as the dependent variable, while previous demand values were used to capture time dependence in the series. Seasonal components were included to reflect repeated patterns over time, and temperature, holiday indicator, and day-of-week dummy variables were added as exogenous variables (Alharbi and Csala, 2022).

In addition to SARIMAX, SARIMA model was also estimated as a benchmark time-series model. Unlike SARIMAX, SARIMA does not include exogenous variables and relies only on the historical electricity demand series and its seasonal structure. This makes SARIMA useful for examining how much forecasting information can be captured from past demand values alone, without the help of external predictors.

A general SARIMAX model can be written as (Hyndman and Koehler, 2006):

$$\Phi_P(B^S) \phi_p(B) (1 - B)^d (1 - B^S)^D y_t = \Theta_Q(B^S) \theta_q(B) \varepsilon_t + \beta^\top \mathbf{X}_t \quad (4.2.1)$$

A general SARIMA model can be written as

$$\Phi_P(B^S) \phi_p(B) (1 - B)^d (1 - B^S)^D y_t = \Theta_Q(B^S) \theta_q(B) \varepsilon_t \quad (4.2.2)$$

where y_t represents electricity demand at time t ; $\mathbf{X}_t$ represents the exogenous variable or set of exogenous variables at time t ; ε_t represents the random error term at time t ; The term β represents the coefficient associated with the exogenous variable, measuring the influence of external factors such as temperature on electricity demand (Alharbi and Csala, 2022).

The backshift operator B is defined as $By_t = y_{t-1}$. The non-seasonal autoregressive component $\phi_p(B)$, of order p , captures the relationship between current demand

and its previous values. The non-seasonal moving average component $\theta_q(B)$, of order q , captures the effect of past forecast errors on current demand. The term $(1 - B)^d$ is used to address non-stationarity, where d is the number of non-seasonal differences.

The seasonal component of the model is represented by $\Phi_P(B^S)$, $\Theta_Q(B^S)$, and $(1 - B^S)^D$. The parameter P is the seasonal autoregressive order, Q is the seasonal moving average order, D is the seasonal differencing order, and S is the seasonal period. These terms allow the model to capture recurring patterns such as daily or weekly cycles in electricity demand.

For the present research, y_t refers to the electricity demand observed at time t , while X_t represents the set of exogenous predictors included in the model, namely temperature, holiday indicator, and day-of-week dummy variables. These exogenous variables were included because electricity demand is influenced not only by its historical patterns, but also by weather conditions, holiday effects, and systematic differences across days of the week. Before estimating the model, the data were arranged in chronological order and divided into training and testing sets. The training set was used to estimate the model parameters, while the testing set was used to evaluate forecasting performance for the day-ahead prediction task. Model orders were selected by examining the characteristics of the time series and comparing candidate specifications using model selection criteria such as AIC. This specification is consistent with previous studies showing that autoregressive models remain useful in short-term electricity demand forecasting, including applications in Australia (Alharbi and Csala, 2022; Al-Musaylh et al., 2018).

4.2.3 Training Strategy

The dataset was divided into 80% for training and 20% for testing, and the data from the last day were removed. After model estimation, forecasting performance was evaluated using a rolling one-step-ahead forecasting strategy on the test set. Model orders were selected through a grid search based on the Akaike Information Criterion (AIC). The search considered non-seasonal parameters over the ranges $p \in \{0, 1, 2\}$, $d \in \{0, 1\}$, and $q \in \{0, 1, 2\}$, and seasonal parameters over $P \in \{0, 1\}$, $D \in \{0, 1\}$, and $Q \in \{0, 1\}$, with the seasonal period fixed at $S = 7$ to reflect the weekly pattern in daily electricity demand. Each candidate SARIMAX model was fitted to the training set together with its exogenous variables, and the specification with the lowest AIC was retained. Based on this procedure, SARIMAX(1, 1, 2) $\times$ (0, 0, 1)₇ was identified as the preferred specification.

For this study, SARIMA was estimated to use the same training and testing split as SARIMAX so that the two models could be compared under a consistent forecasting setting. Model orders were selected through a grid search based on the Akaike Information Criterion. Based on this procedure, the best SARIMA specification was identified as SARIMA (2,0,2) $\times$ (0,1,1,7).

4.2.4 Results

The forecasting performance of the time-series model was evaluated by using four different standard error measures: 4.1.5–4.1.8: MAE, RMSE, MAPE and R^2 . MAPE should be interpreted with caution when actual values are very small (Hyndman and Koehler, 2006).

To compare the role of external variables in forecasting, both SARIMA and SARIMAX were evaluated in this study. SARIMA was used as a benchmark model based only on historical demand and seasonal structure, while SARIMAX extended this framework by including exogenous variables. Table X summarizes the forecasting results of the two models.

Table 4.2: Forecasting performance of SARIMA and SARIMAX

Model	Order	Seasonal Order	MAE	RMSE	MAPE	R^2
SARIMA	(2,0,2)	(0,1,1,7)	14267.220	19808.805	3.779%	0.729
SARIMAX	(1,1,2)	(0,0,1,7)	11325.733	15753.188	2.991%	0.828

As shown in Table, both models were able to provide useful day-ahead forecasts, but the SARIMAX model produced lower error values than SARIMA across all three measures. It shows that including external variables can improve forecasting performance. The lower MAE and RMSE indicate that SARIMAX generated smaller average errors and fewer large deviations, while the lower MAPE suggests better relative accuracy in percentage terms. Therefore, the comparison shows that exogenous variables such as temperature, holiday effects, and day-of-week indicators contributed additional predictive information in this study. This suggests that historical demand alone does not fully explain day-ahead changes in electricity demand, and that external drivers provide additional predictive information. (Hyndman and Koehler, 2006; Nti et al., 2020).

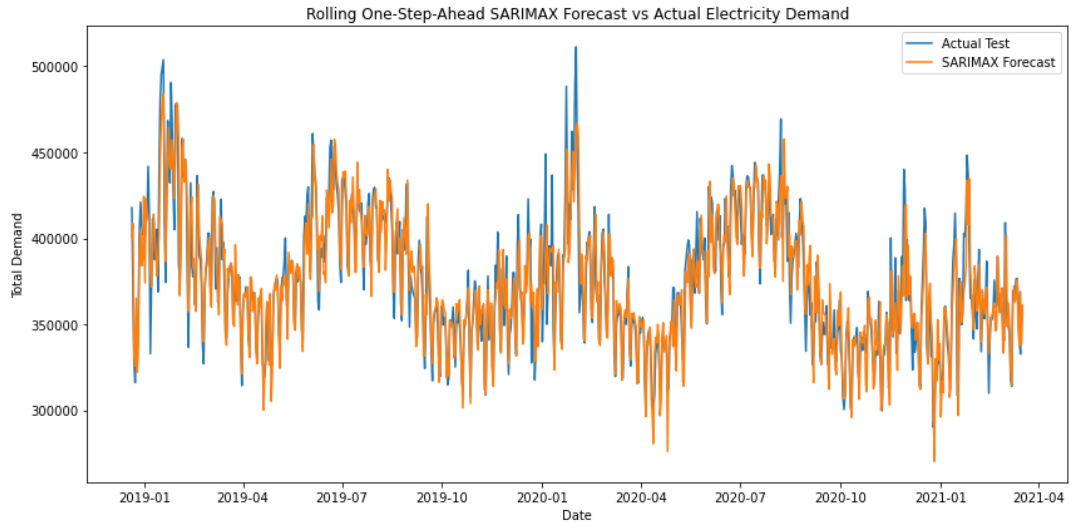


Figure 4.1: Actual rolling SARIMAX-predicted electricity demand in the test period (January 2019 – April 2021) for the SARIMAX(1, 1, 2) $\times$ (0, 0, 1, 7) model.

Figure 4.1 compares the predicted demand series with the actual observed values in the test period. The figure shows that the predicted values generally follow the movement of the actual demand series, although the forecasts are smoother and do not fully capture some short-term fluctuations and peaks. This suggests that while

the SARIMAX model is effective in modelling the broader temporal pattern, the model performance could still be improved, especially when demand changes more sharply (Nti et al., 2020).

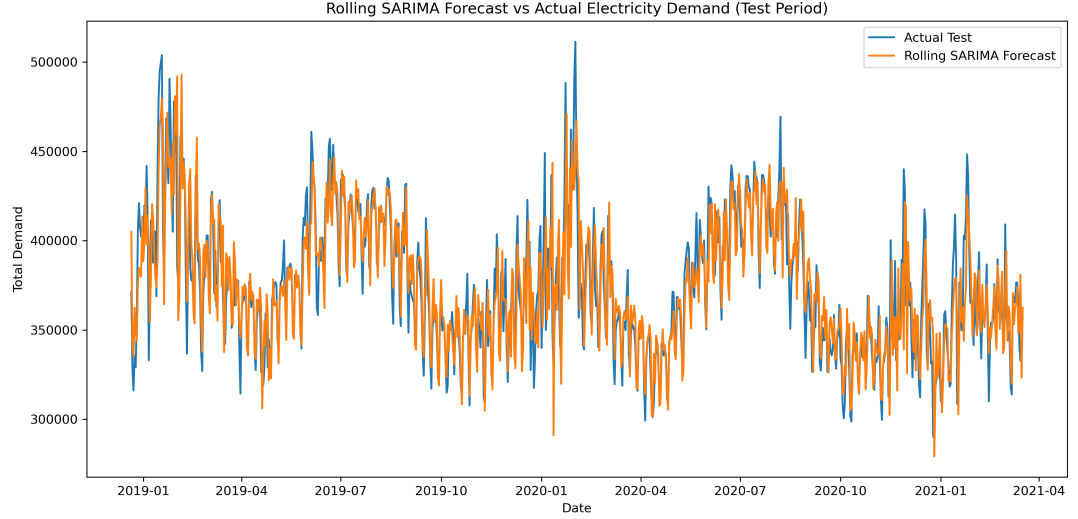


Figure 4.2: Actual versus rolling SARIMA-predicted electricity demand in the test period (January 2019 – April 2021) for the $SARIMA(2, 0, 2) \times (0, 1, 1, 7)$ model.

Figure 4.2 shows that the rolling SARIMA forecast was able to track the general movement of electricity demand during the test period. However, SARIMA still produced larger forecasting errors than SARIMAX, which suggests that a model using only past demand values and seasonal structure may be less effective than a model that also includes exogenous predictors.

The comparison between SARIMA and SARIMAX shows that adding external variables helped improve the forecasting results in this study. Both models could follow the main trend and seasonal pattern of electricity demand, but SARIMAX gave smaller errors and a higher R^2 value, which means it matched the actual data better. This suggests that day-ahead electricity demand depends not only on past demand values, but also on factors such as temperature, holidays, and weekday differences. However, both models still produced smoother forecasts than the real series, especially when demand changed suddenly or reached a peak. So, although SARIMAX performed better in this study, there is still some space to improve the model further.

4.2.5 Strengths and Limitations

SARIMA is a useful benchmark in this study because it models electricity demand by using only its own historical values and seasonal structure. This makes it relatively simple and interpretable compared with more complex forecasting approaches. Since electricity demand often shows repeated weekly and seasonal patterns, SARIMA is able to capture an important part of the temporal structure in the series. In this study, the SARIMA results show that a model based only on past demand and seasonality can already provide a reasonable level of forecasting accuracy.

Compared with SARIMA, SARIMAX model offers a more flexible framework because it can incorporate both time-series structure and external variables. The model can capture temporal dependence and seasonal patterns in electricity demand, while also incorporating exogenous variables such as temperature, holiday effects, and day-of-week differences. This makes SARIMAX a useful model for day-ahead electricity demand forecasting, and it also an appropriate benchmark for comparison with other forecasting approaches (Czapaj et al., 2022; Nti et al., 2020).

The SARIMAX model also has some limitations. Firstly, the forecasting performance depends on the correct specification of model orders and seasonal structure. Secondly, although the model can track the main pattern of electricity demand, the forecasts are smoother than the actual series and may not fully capture sudden fluctuations or demand peaks. Thirdly, SARIMAX mainly reflects linear relationships and may be less effective when the data contains more complex nonlinear behavior. Therefore, while SARIMAX provides a solid and interpretable baseline, further improvements may be achieved by comparing it with more flexible machine learning or deep learning models.

4.3 Machine Learning

4.3.1 Model Selection

Based on the regression and time series analyses, this section employs a two-group comparison framework to model and forecast daily electricity demand in NSW. Mansour et al. (2026) showed that both XGBoost and LightGBM performed well in hourly-scale forecasts. Combining these findings with those from the regression section, the first group (Group A) consists of gradient boosting variants based on ensemble learning—Gradient Boosting (GB), XGBoost and LightGBM. Following the hybrid modelling approach proposed by Hassanpour et al. (2025), and combining existing findings from the time series analysis, the second group (Group B) constructs two-stage hybrid models based on SARIMAX and the same three machine learning algorithms. To ensure that each model achieves its best performance, all ML models in both groups are tuned by Optuna’s Tree-structured Parzen Estimator (TPE) Bayesian search.

This setup is designed to answer the following research questions:

1. Are machine learning models sufficient to capture the seasonality of electricity demand?
2. Can hybrid models achieve better forecasting results?

4.3.2 Model Specification

ML Models

XGBoost (eXtreme Gradient Boosting) was proposed by Chen and Guestrin (2016). Building upon the standard gradient boosting framework, XGBoost introduces a second-order Taylor expansion to achieve a more accurate approximation of the objective function. In addition, L_1 (Lasso) and L_2 (Ridge) regularisation terms are incorporated during tree growth to effectively control model complexity:

$$\mathcal{L}^{(t)} = \sum_{i=1}^n \left[g_i f_t(\mathbf{x}_i) + \frac{1}{2} h_i f_t^2(\mathbf{x}_i) \right] + \Omega(f_t), \quad (4.3.1)$$

where g_i and h_i denote the first- and second-order derivatives of the loss function with respect to the predicted value, respectively, and $\Omega(f_t) = \gamma T + \frac{1}{2} \lambda \|w\|^2$ represents the regularisation term, in which T is the number of leaf nodes and λ is the L_2 regularisation coefficient. This design enables XGBoost to achieve superior generalisation ability and computational efficiency, particularly when handling sparse features and large-scale datasets. Early stopping was implemented using an internal validation set with `early_stopping_rounds = 50` to enhance the model’s generalisation performance.

LightGBM (Light Gradient-Boosting Machine) was proposed by Ke et al. (2017) from Microsoft Research. To address the issue of training efficiency with large-scale data, it introduced two key innovations:

- **Gradient-based One-Side Sampling (GOSS):** This method prioritises retaining samples with large gradient magnitudes (which contribute more information) and randomly samples those with small gradients, thereby reducing the data volume whilst maintaining the accuracy of distribution estimation.
- **Exclusive Feature Bundling (EFB):** This technique merges mutually exclusive sparse features into a single feature, thereby reducing the feature dimension and minimising memory usage and computational cost.

Furthermore, LightGBM employs a leaf-wise tree-growing strategy. Compared to XGBoost’s level-wise strategy, this achieves lower loss function values for the same number of iterations; however, control over `num_leaves` is more critical, as an excessively high value can easily lead to overfitting. In this study, we set `num_leaves = 63`, `feature_fraction = 0.8` and `bagging_fraction = 0.8`, and enabled early stopping (50 rounds) to balance model complexity with the risk of overfitting.

Hybrid SARIMAX–Machine Learning Model

This paper employs a two-stage hybrid forecasting framework based on residual correction. In the first stage, the SARIMAX model specification identified in the time series section—order (1, 1, 2) and seasonal order (0, 0, 1, 7), selected via AIC grid search—is adopted as the baseline. Mean temperature (T_{mean}), a public holiday indicator (`IS_HOLIDAY`), and six day-of-week dummy variables (`WD_2–WD_7`) serve as exogenous regressors, capturing the linear effects of temperature, public holidays, and intra-week demand variation. For computational efficiency within the two-stage framework, the baseline forecast on the test set is obtained via `apply()`, which propagates the estimated state space using the trained parameters without re-fitting the model at each step, rather than the full rolling re-estimation used in the time series section.

In the second stage, the residuals generated by the first-stage model are treated as the target variable. Temperature features (T_{max} , T_{min} , and T_{mean}) and calendar features (`WEEKDAY`, `IS_HOLIDAY`, `YEAR`, `MONTH`, `DAY`, `DAYOFWEEK`, `DAYOFYEAR`, `WEEKOFYEAR`) are then used to train Gradient Boosting, XGBoost and LightGBM

models separately for residual correction. Although `IS_HOLIDAY` and `WEEKDAY` are also included in the SARIMAX exogenous specification, their presence in the ML stage allows the residual-correction model to capture any non-linear interactions involving these variables that the linear SARIMAX framework cannot represent. Lag demand features (`DEMAND_LAG1/LAG7`) are deliberately excluded from the second stage: since the SARIMAX baseline already encodes the autoregressive structure of the demand series, including lag features in the residual-correction model would duplicate this information and introduce data leakage.

The final forecast is obtained by summing the baseline forecast from the SARIMAX model with the residual correction term provided by the machine learning model:

$$\hat{y}_t = \hat{y}_{\text{SARIMAX},t} + \hat{e}_t, \quad \hat{e}_t = f_{\text{ML}}(Z_t), \quad (4.3.2)$$

where $\hat{y}_{\text{SARIMAX},t}$ denotes the baseline forecast generated by SARIMAX, $\hat{e}_t$ denotes the predicted residual from the machine learning model, and Z_t represents the set of temperature and calendar features (excluding lag demand) used in the second stage. This phased design allows SARIMAX to model trends, seasonality, and the linear effects of temperature and calendar variables, while the machine learning models further capture non-linear temperature–demand relationships and higher-order calendar interactions that elude SARIMAX’s linear framework.

4.3.3 Training Strategy

Feature Engineering

The target variable is `TOTAL_DEMAND` throughout. For the Group A models, the full feature set includes: temperature variables (`T_MAX`, `T_MIN`, `T_MEAN`), calendar variables (`WEEKDAY`, `IS_HOLIDAY`, `YEAR`, `MONTH`, `DAY`, `DAYOFWEEK`, `DAYOFYEAR`, `WEEKOFYEAR`), and lag demand variables (`DEMAND_LAG1`, `DEMAND_LAG7`).

In Group B, feature assignment follows the functional division of the two-stage framework. The SARIMAX model receives mean temperature (`T_MEAN`), a holiday indicator (`IS_HOLIDAY`), and six day-of-week dummy variables (`WD_2–WD_7`) as exogenous regressors. The residual-correction ML model in the second stage is supplied with all three temperature variables (`T_MAX`, `T_MIN`, `T_MEAN`) and the remaining calendar variables, but excludes the lag demand features (`DEMAND_LAG1/LAG7`). This design prevents data leakage from the autoregressive structure already modelled by SARIMAX, while allowing the ML model to capture the non-linear relationship.

Data Splitting and Evaluation

This section follows the same approach as the regression section, splitting the dataset into a training set and a test set in an 80:20 ratio, without random shuffling. Within the training set, the final 10% is further set aside as a validation set, dedicated to determining early stopping for XGBoost and LightGBM; Gradient Boosting controls the stopping condition by setting `n_iter_no_change = 50`, meaning that training terminates automatically if the model fails to achieve a significant improvement in performance over 50 consecutive iterations. The test set is used only for final evaluation once all models have been finalised and is not involved in any hyperparameter tuning throughout the process.

The forecast evaluation uses the same error measures defined in Equations 4.1.5–4.1.8: MAE, RMSE, MAPE and R^2 .

Model Training and Hyperparameter Tuning

Table 4.3 shows the optimal hyperparameter configurations identified by Optuna for all ML models. For Group A models, Optuna searched over 60 trials (GB) or 100 trials (XGBoost and LightGBM) using 5-fold `TimeSeriesSplit` cross-validation with a gap of 7 days between the training fold and the validation fold, minimising the cross-validated MAE. The same search procedure was applied to each Group B residual-correction model, using the SARIMAX training residuals as the target variable. The SARIMAX baseline hyperparameters (order and seasonal order) remain as determined in the time series section and were not re-tuned.

Table 4.3: Optimal hyperparameter configurations identified by Optuna TPE for all models.

Model	Parameter	Value
<i>Group A: Standalone Machine Learning Models</i>		
Gradient Boosting	<code>n_estimators</code>	701
	<code>learning_rate</code>	0.0443
	<code>max_depth</code>	4
	<code>subsample</code>	0.836
	<code>min_samples_leaf</code>	11
XGBoost	<code>n_estimators</code>	695
	<code>learning_rate</code>	0.0217
	<code>max_depth</code>	5
	<code>subsample</code>	0.737
	<code>colsample_bytree</code>	0.834
	<code>reg_alpha</code>	0.00483
	<code>reg_lambda</code>	0.0160
LightGBM	<code>n_estimators</code>	836
	<code>learning_rate</code>	0.0114
	<code>num_leaves</code>	48
	<code>feature_fraction</code>	0.659
	<code>bagging_fraction</code>	0.600
	<code>bagging_freq</code>	5
	<code>min_child_samples</code>	10
	<code>reg_alpha</code>	0.758
	<code>reg_lambda</code>	0.0758
<i>Group B: Hybrid SARIMAX + ML Models</i>		
SARIMAX (baseline)	order (p, d, q)	(1, 1, 2)
	seasonal order (P, D, Q, s)	(0, 0, 1, 7)
	exogenous variables	T_MEAN, IS_HOLIDAY, WD_2--WD_7
	<code>enforce_stationarity</code>	False

Continued on next page

Table 4.3 (continued)

Model	Parameter	Value
	maxiter	500
SARIMAX + GB (ML stage)	n_estimators	718
	learning_rate	0.00624
	max_depth	7
	subsample	0.657
	min_samples_leaf	22
SARIMAX + XGBoost (ML stage)	n_estimators	346
	learning_rate	0.0196
	max_depth	4
	subsample	0.890
	colsample_bytree	0.681
	reg_alpha	0.0293
	reg_lambda	0.00222
SARIMAX + Light-GBM (ML stage)	n_estimators	212
	learning_rate	0.00897
	num_leaves	66
	feature_fraction	0.702
	bagging_fraction	0.812
	bagging_freq	5
	min_child_samples	21
	reg_alpha	0.0543
	reg_lambda	0.0139

4.3.4 Results

Table 4.4 reports the out-of-sample evaluation metrics for all six Optuna-tuned models on the test set. Across both groups, the standalone machine learning models in Group A perform better than the hybrid models in Group B.

Within Group A, Gradient Boosting achieves the best MAE of 8,776 MWh and MAPE of 2.37%, LightGBM attains the best RMSE of 11,546 MWh and the highest R^2 of 0.9079. These differences are negligibly small in absolute terms, suggesting that at this data scale all three gradient boosting variants are nearly equivalent once each is properly calibrated. Within Group B, SARIMAX + XGBoost achieves the best MAE of 9,792 MWh and R^2 of 0.7916 among hybrid models.

Table 4.4: Out-of-sample evaluation results for all Optuna-tuned models on the test set. MAE and RMSE are in MWh; the best result in each column is shown in **bold**.

Group	Model	MAE (MWh)	RMSE (MWh)	MAPE (%)	R^2
Group A	Gradient Boosting	8,776	11,591	2.37	0.9072
	XGBoost	8,795	11,585	2.38	0.9073
	LightGBM	8,847	11,546	2.40	0.9079
Group B	SARIMAX + XGBoost	9,792	17,371	2.64	0.7916
	SARIMAX + LightGBM	9,902	17,560	2.66	0.7870
	SARIMAX + GB	9,914	17,562	2.67	0.7870

Although Shiblee and Koukaras (2025) and Inala and Manojkumar (2025) both found LightGBM to perform best in their respective studies, the differences among the three Group A models in this study are too small to identify a clear winner: the MAE spread is only 71 MWh (less than 0.9% of the best value), and the ranking differs depending on the metric chosen.

4.3.5 Limitations

Relatively small sample size. Due to the daily frequency of the data, the training set comprises only a few thousand rows, which is far smaller than the scale typically seen in studies using minute- or hour-level data. This limits the performance of XGBoost and LightGBM.

Error propagation in the two-stage hybrid design. The forecast errors from the SARIMAX model are directly carried over into the residuals of the second stage. More significantly, SARIMAX employs linear modelling for the temperature effect, yet the scatter plot from the exploratory data analysis reveals a clear U-shaped non-linear relationship between temperature and demand (heating at low temperatures, cooling at high temperatures). This implies that a considerable amount of non-linear signal remains in the residuals. Although the second-stage ML model includes all three temperature variables ($T_{\max}$, $T_{\min}$, T_{mean}) and is in principle capable of capturing this non-linearity, the residual signal it receives has already been partially distorted by the SARIMAX linear approximation. The ML model is therefore correcting a linearised residual rather than modelling the raw non-linear temperature–demand relationship directly, which limits the effectiveness of the residual correction and remains a core structural reason why Group B performs overall worse than Group A.

Applicability of weather features in operational forecasting. The temperature data used in this study represents the *actual* temperature on the day of the forecast. However, in operational day-ahead forecasting, actual temperatures for future days are unavailable; real-time weather forecast data would therefore need to be used as a substitute, introducing an additional source of uncertainty.

4.4 Deep Learning Modeling Approach

4.4.1 Model Selection

There are multiple moving parts when it comes to Electricity demand forecasting. There is a non-linear relationship among most variables, but there is also temporal dependence, implying that the data cannot be treated as independent values either. For instance, if we look at the Exploratory Data Analysis(EDA) of the data, we realise that lagged variables like `DEMAND_LAG1` and `DEMAND_LAG7` show clear auto-correlation, which tells us that demand strongly depends on its past values.

Moreover, it's not only the past values. If we take calendar effects and temperature into consideration, sometimes their impact on demand is huge, and other times it doesn't cause any change. While they may have a weak correlation, there is a meaningful influence. This happens due to multiple interacting factors, such as temperature and whether it's a weekday or a holiday. This renders linear models insufficient.

Due to this, deep learning models make sense here. They are capable of handling complex patterns between variables, and at the same time, don't expect us to provide the exact relationship between fields or variables. In particular, this motivates the use of both ANN and LSTM models to capture nonlinear relationships and temporal dependencies.

The ANN model is used as the baseline for this approach. It is capable of capturing many nonlinear relationships between the variables and the electricity demand. However, it does not account for the temporal order of the data, because there is no memory aspect like LSTM. Instead, time dependence is indirectly added through the lagged demand variables.

The LSTM model is used as the main deep learning model for this approach. This is due to the fact that LSTM uses memory cells, which help retain information from previous steps. This makes it very suited for this time series setting, where the current depends on the past values. LSTM uses this to learn sequential patterns in the data. Furthermore, in EDA, there is a strong temporal structure. The LSTM network is expected to perform better than the ANN network.

This comparison is very important because it allows us to evaluate two different approaches to handling temporal dependence. ANN approaches it using engineered lag features, while LSTM learns temporal relationships directly from the sequential input data. This allows us to assess whether temporal dependence is better captured through explicit feature engineering or through sequence-based learning.

4.4.2 Model Specification

Artificial neural networks(ANNs) were modelled after the human brain(Tenny et al., 2023). ANNs consist of layers of interconnected neurons, where each neuron applies a weighted transformation on its input, and then it is passed through a nonlinear activation function. Then it goes through training where the network learns to predict the output function using the input, despite their complex relationship.

Because ANNs don't account for temporal dependencies in sequential data (Alzubaidi et al., 2021), to overcome this limitation, recurrent neural networks(RNNS) were developed. They allow previous time steps to affect the later time steps, and information is passed on. LSTM is a specialised RNN that was designed to overcome the

issue of vanishing gradients (Hochreiter and Schmidhuber, 1997) and their inability to let go of long-term dependencies on previous patterns.

Specifically, LSTM networks have many gating mechanisms, like input, forget and output gates, which, along with the memory cells mentioned previously, regulate how information flows throughout the entirety of the network. It's through these components that the model decides what information to retain and what information to discard. Hence, when there are time series problems like electricity forecasting, we choose LSTM because it utilises information from past observations while focusing on the current observation as well.

Artificial Neural Network (ANN)

The ANN model consists of an input layer, two hidden layers and an output layer. It also consists of multiple interconnected layers that further transform input data into future predictions. The first hidden layer has 64 neurons. Then it is followed by a dropout layer that reduces overfitting. Then a second hidden layer exists containing 32 neurons. All the hidden layers of the ANN use the ReLU activation function.

The output layer of the ANN model contains a single neuron, which outputs the predicted electricity demand for the day, implying that there is only one output. The ANN learns all non-linear relationships between input features such as temperature, calendar variables, and lagged demand. Though unlike LSTM models, it is unable to capture temporal dependencies, as each observation is treated independently. The ANN was trained using the Adam optimiser. We further used mean squared error as the loss function. We also used early stopping, which is based on the validation loss, to reduce overfitting and improve generalisation performance.

Long Short-Term Memory (LSTM)

The LSTM model is designed to find patterns that happen over time using information that comes in one after the other (sequentially). This model has two LSTM layers stacked on top of each other. Then it is followed by a dense output layer. The first LSTM layer has 64 units. It processes the input sequence, and then it sends that information to the next layer. The second LSTM layer has 32 units. It tries to understand the patterns that happen over time, and then it produces a simple summary representation. There is also a layer between the LSTM layers called the Dropout layer, which helps reduce overfitting and improve generalisation.

The information that goes into the LSTM model is a sequence of observations that happened over the seven days. Each timestep has information like what the temperature was, what day it was, and how much of something was needed and the demand values. The model was trained using the Adam optimiser and Mean Squared Error (MSE) loss function. Early stopping is also applied based on the degree of validation. This architecture allows the model to learn time-dependent relationships effectively from sequential data.

4.4.3 Training Strategy

The main target variable for the models was TOTAL_DEMAND. Based on EDA, the main explanatory variables are temperature, calendar features and lagged demand variables. Moreover, to avoid multicollinearity, we kept only T_MEAN. The fact that T_MAX and T_MIN were highly correlated with T_MEAN meant that including all of them was not beneficial.

We used WEEKDAY and IS_HOLIDAY to capture the calendar effects. We also included DEMAND_LAG1 and DEMAND_LAG7 in the ANN model to account for temporal dependence.

The ANN model required the data to be structured in tabular format, where each row represented exactly one day. The LSTM model required a different structure. So the data was converted into sequences, and a rolling window was used to help predict the demand for the upcoming day.

Additionally, all input variables were scaled using Min-Max normalisation before training. Then the dataset was split into training and testing sets using the 80-20 split, where 80% of the data is used for training and the rest 20% is for testing. Furthermore, the split was done in chronological order to maintain the temporal order of the data and avoid data leakage. These steps were taken to ensure all features are on a comparable scale and improve model convergence during the training phase.

To further evaluate model performance, we used the following evaluation metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and the coefficient of determination (R^2).

MAE gives the average size of the errors, which makes it easy to understand how far off the centre the predictions are on average. RMSE is slightly different, since it penalises larger errors more heavily, implying that values at the end of the scale have a large impact, as the values are squared and then the root mean square is taken. It becomes a very useful metric when we want to see how the model performs on extreme demand values.

MAPE expresses the error in percentage terms, which helps in interpreting performance relative to the demand rather than absolute values. This is particularly helpful in our case, like electricity demand, where the values can be very different across time. R^2 was also used to measure how well the model explains the variation in TOTAL_DEMAND. A higher R^2 indicates that the model is capturing more of the underlying structure in the data.

When all the metrics are used together, we get a much more complete picture. It also gets a lot easier to compare models and decide which one works more efficiently. Some focus on average error, while some look at overall fit or the impact of extreme values. Utilising all the metrics gives us a lot more perspective on the performance of the model.

4.4.4 Results

Table 4.5 summarises the out-of-sample performance of the ANN and LSTM models on the test set. The forecast evaluation uses the same error measures defined in Equations 4.1.5–4.1.8: MAE, RMSE, MAPE and R^2 .

Table 4.5: Performance comparison of ANN and LSTM models on the test set. MAE and RMSE are in MWh; the best result in each column is shown in **bold**.

Model	MAE (MWh)	RMSE (MWh)	MAPE (%)	R^2
ANN	11,446.573	19,064.764	9.049	0.7743
LSTM	12,883.740	21,681.029	9.537	0.7081

The results show us that the ANN model clearly outperformed the LSTM model across all four evaluation metrics. The ANN has achieved lower Mean absolute error (MAE), Root Mean Square error (RMSE) and Mean absolute Percentage error (MAPE). It also achieved a higher R^2 . All of this indicated that it had a better overall predictive performance.

Why did ANN perform better than LSTM?

1. The ANN was given access to lagged demand variables (DEMAND_LAG1, DEMAND_LAG7). They directly capture the temporal dependence, but LSTM had no direct access to these features and had to learn them from the sequences given to it, which were only 7 days long. (Weekly electricity forecasting) This provided it with a relatively limited context when it came to predicting the next week.

2. The dataset has a relatively small size (4000 observations). Complex models like LSTM require more data, and it was simply not enough for it to learn sequential patterns well enough. The LSTM has 30k parameters when compared to the 2.5k in the ANN. The LSTM is harder to train effectively and requires more data to perform well.

3. The ANN benefits from explicit feature engineering, including temperature, calendar variables, and lagged demand, which are strong predictors of electricity demand. The LSTM, although it includes TOTAL_DEMAND in sequences, does not use an explicit lag structure. This makes it harder for it to learn clear temporal relationships, as well as ANN.

4. The problem is mainly driven by short-term dependencies (daily and weekly patterns). This is already well captured by the lag features. This erases the advantage of using a sequence-based model such as an LSTM. Larger sequences can help train LSTM better.

As a result, the ANN achieved better performance across all evaluation metrics.

Figure 4.3 presents a comparison of actual and predicted electricity demand for both models.

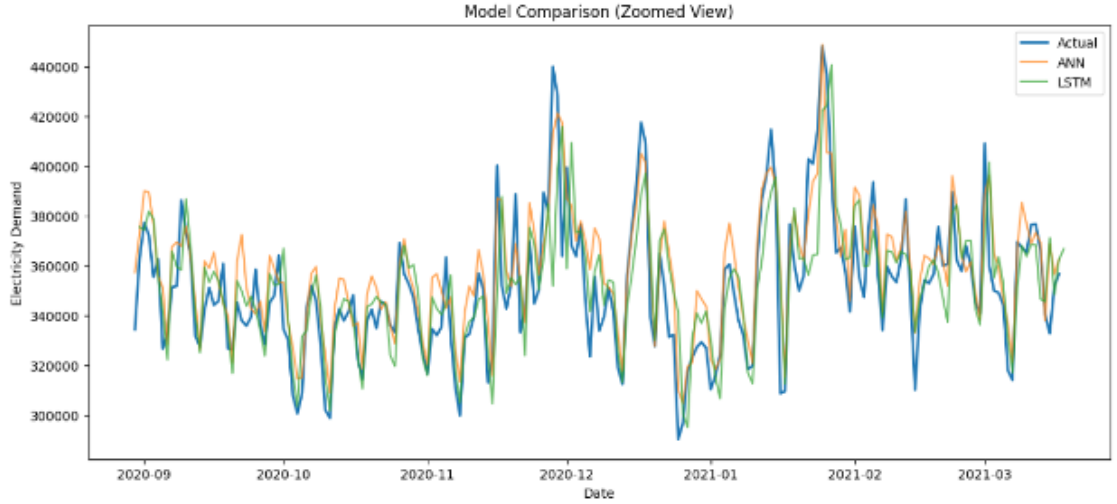


Figure 4.3: Actual vs. predicted electricity demand: ANN and LSTM models on the test set (zoomed view).

The ANN model closely follows the actual demand series, capturing both the overall trend and all the short-term fluctuations. The LSTM model is able to replicate the general pattern, but it exhibits way larger deviations, particularly during peak demand periods, and these findings are very consistent with multiple previous studies. While deep learning models are really powerful, their performance depends heavily on the data size and feature representation. (Nti et al., 2020).

4.4.5 Limitations

The following are the limitations of the model:

Limited benefit of LSTM over ANN.

The ANN model already includes lagged demand variables from the dataset. This captures a large part of the time structure on its own. So, the added benefit of using a sequence-based model like LSTM is limited and not always necessary because LSTM has to individually calculate the lagged demand variables, but ANN gets them as part of the dataset.

Sensitivity to hyperparameters.

Deep learning models require very careful hyperparameter tuning. If the settings are not optimal, as in this case, performance can suffer and more complex models may perform worse.

CHAPTER 5

Conclusion

This study compared regression, time series, machine learning, and deep learning approaches for day-ahead electricity demand forecasting in New South Wales using historical electricity demand, temperature variables, and calendar information. Table 5.1 presents the overall performance comparison of all models on the test set.

Table 5.1: Overall performance comparison of all models on the test set. MAE and RMSE are in MWh; the best result in each column is shown in **bold**. The AEMO official forecast is included as a reference benchmark only and is excluded from bold selection.

Model	MAE (MWh)	RMSE (MWh)	MAPE (%)	R^2
AEMO Official Forecast	808.459	1,256.898	0.205	0.9989
Gradient Boosting	8,776.000	11,591.000	2.370	0.9072
LightGBM	8,847.000	11,546.000	2.400	0.9079
XGBoost	8,795.000	11,585.000	2.380	0.9073
Random Forest	10,868.000	14,567.000	2.950	0.8535
SARIMAX	11,325.733	15,753.188	2.991	0.8280
SARIMAX + XGBoost	9,792.000	17,371.000	2.640	0.7916
ANN	11,446.573	19,064.764	9.049	0.7743
LSTM	12,883.740	21,681.029	9.537	0.7081
Ridge	14,127.803	18,573.265	3.728	0.7617
Linear Regression	14,127.280	18,574.480	3.727	0.7617
Lasso	14,139.902	18,581.039	3.732	0.7615
SARIMA	14,267.220	19,808.805	3.779	0.7293

Looking first at the regression models, Random Forest was clearly the strongest of the group. It performed better than Linear Regression, Ridge, and Lasso, which suggests that the predictors do contain useful information for forecasting NSW electricity demand. At the same time, the weaker results from the linear models imply that the relationships in the data are not mainly simple or linear. Regularisation helped with the high correlation among the temperature variables, but that alone was not enough to close the gap. A more flexible model was still needed to capture nonlinear effects and variable interactions.

The time-series models were still useful, even if they were not the strongest overall. Both SARIMA and SARIMAX were able to track the general movement of the series and pick up the repeated seasonal pattern in electricity demand. That fits what was already seen in the exploratory analysis, where lag effects and weekly repetition were quite obvious. SARIMAX did better than SARIMA, so adding outside information clearly helped. The temperature and calendar variables were not just extra controls in the model; they improved the forecasts in a noticeable way. Even so, the best time-series results were still weaker than the best machine learning results, which suggests that the NSW demand series contains more than seasonal linear structure alone.

Among the machine learning models, the differences were small enough that they should not be overstated. Gradient Boosting produced the lowest MAE and MAPE, so it was the most accurate on average across the test period. LightGBM had the lowest RMSE and the highest R^2 , which suggests that it dealt slightly better with larger misses and gave the strongest overall fit. XGBoost was very close to both. In practical terms, these three models were operating at almost the same level, even though each one had a slightly different strength depending on the metric being used.

The deep learning results were more straightforward. ANN outperformed LSTM on every evaluation metric, so in this dataset the more complex sequence model did not bring an advantage. One likely explanation is that much of the predictable structure in electricity demand had already been captured through lagged demand variables and the selected inputs. In that case, the LSTM architecture may simply have had less extra structure left to learn. Another possible reason is the size of the dataset. For a daily series of this scale, the data may be enough for ANN, but still limited for a more complex model such as LSTM.

Put all the results together, the tree-based machine learning models gave the best predictive results in this study. The gap within the top three machine learning models was small. The MAE difference between Gradient Boosting, XGBoost, and LightGBM was less than 1%, which implies that changing from one strong boosting model to another did not make much practical difference. Compared with the official prediction results of AEMO, the results of this paper are not enough to reach the level of industrial use. The gap here comes from the rich feature input of the official model.

This point matters for how the results should be interpreted. In the NSW day-ahead setting, once the model is flexible enough to handle nonlinear relationships, further gains may depend less on switching algorithms and more on improving the information fed into the model. In other words, feature design may matter more than model architecture once a reasonably strong method is already in place. For practical forecasting work, that is an important result. It suggests that collecting better weather inputs, refining calendar effects, or adding other useful external variables may be more valuable than repeatedly changing between similar high-performing models.

The results also have some practical value for decision-making. If the priority is to improve average forecast accuracy in routine day-ahead planning or bidding, Gradient Boosting looks slightly more suitable because it achieved the lowest MAE. If the main concern is reducing the risk of larger forecast errors, LightGBM may

be the better choice because it had the lowest RMSE. So even when the overall performance of the best models is quite close, the final decision can still depend on the operational objective.

There are also some limits to these results. The predictor set is still fairly narrow, since the models mainly use historical demand, temperature, weekdays, months, and holidays. Other factors that could matter for electricity demand, such as humidity, wind, economic conditions, policy change, market conditions, or unexpected shocks, were not included. The larger errors near the end of the sample, especially around 2020, suggest that these inputs do not explain all unusual movements very well. The evaluation is also based on one dataset and one chronological 80:20 train–test split, so firmer evidence would come from checking performance across several rolling forecast windows. Another limitation is that the study only looks at point forecasts and does not report prediction intervals or other measures of uncertainty, even though those can be useful in practice. Finally, although the models were tuned in a sensible way, the reported results should still be read as what was achieved under the current design, not as the absolute best possible outcome for each model class.

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AI Usage Statement

We acknowledge the assistance provided by ChatGPT (OpenAI, GPT-5, 2026) in the preparation of this thesis. We used AI to assist with the standardisation of technical terminology and academic wording ensuring that technical expressions were presented in an academically appropriate manner. The code used in this thesis was our own contribution, but when we encountered problems related to Python syntax or errors occurring after running the code, we asked the AI for solutions. We also used AI to convert the content we had written into Latex format. In addition, when reading the literature, we used AI to quickly summarise the content of the literature so that we could review it easily. What's more, we used AI to review our content to ensure our content is logical and accurate.